

Problem 1(b): Integration only over θ ' (here called 'x')

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In[1]:= Integrate[
  r^2 * Exp[-I * k * r * Cos[x]] * Sin[x] * r *
  (Cos[x] * Cos[θ] + Sin[x] * Sin[θ] * Cos[φ - χ]) * Exp[-r/a0], {x, 0, Pi}]
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$$\text{Out[1]} = \frac{2 i e^{-\frac{r}{a_0}} r \cos[\theta] (k r \cos[k r] - \sin[k r])}{k^2} + \frac{e^{-\frac{r}{a_0}} \pi r^2 \text{BesselJ}[1, k r] \cos[\phi - \chi] \sin[\theta]}{k} \text{ if } k r \in \mathbb{R}$$

Problem 1(b): Integration over all whole given volume

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In[2]:= Integrate[
  r^2 * Exp[-I * k * r * Cos[x]] * Sin[x] * r *
  (Cos[x] * Cos[θ] + Sin[x] * Sin[θ] * Cos[φ - χ]) * Exp[-r/a0],
  {x, 0, Pi}, {φ, 0, 2 * Pi}, {r, 0, Infinity}]
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$$\text{Out[2]} = -\frac{32 i a_0^5 k \pi \cos[\theta]}{(1 + a_0^2 k^2)^3}$$

Problem 1(c)

Square the above expression for using in Fermi Golden Rule magnitude (the i 'imaginary' can be simply ignored by multiplying a -1 as we require absolute value only)

$$\begin{aligned} \text{In[19]} &:= -\left(2 \pi / h\right) \left(e * E_0 / (\text{Sqrt}[(\pi * a_0^3 * L^3)]) \right) \frac{32 i a_0^5 k \pi \cos[\theta]}{(1 + a_0^2 k^2)^3} \wedge 2 * \\ &\quad (m * L^3 / (8 \pi^3 h^2) k) \\ \text{Out[19]} &:= \frac{256 a_0^7 e^2 E_0^2 k^3 m \pi^2 \cos[\theta]^2}{h^3 (1 + a_0^2 k^2)^6 \pi^3} \end{aligned}$$

Comment: We have a $\cos(\theta)^2$ dependence for the differential transition rate. The differential is taken with respect to the solid angle differential as discussed in the PS Notes. At $\theta = 90$ we can

see that the probability is 0 which physically makes sense because the Electric Field is then perpendicular to the wave vector.

Problem 1(d)

$$\text{In[20]:= Integrate}\left[\left(\frac{256 a_0^7 e^2 E_0^2 k^3 m \pi^2 \cos[\theta]^2}{h^3 (1 + a_0^2 k^2)^6 \pi^3}\right) \sin[\theta], \{\theta, 0, \pi\}\right]$$

Out[20]=

$$\frac{512 a_0^7 e^2 E_0^2 k^3 m \pi^2}{3 h^3 (1 + a_0^2 k^2)^6 \pi^3}$$

Problem 1(e)